

Units 1-2 Review: Neurons & Memory

Prepared for **Example Student**

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Concepts covered: 2

() bio-neuron-structure

() cog-encoding-storage-retrieval

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Unit 1 — Biological Bases of Behavior

Neuron Structure & the Synapse

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Neurons are nerve cells that transmit information — the basic building blocks of the nervous system. **Glial cells** support them by providing structure, insulation, communication, and waste transport.

Neuron anatomy (signal path: dendrites, soma, axon, terminal buttons):

- **Dendrites:** root-like branches that *receive* signals from other neurons; they grow to form new synaptic connections (learning grows dendritic branches).
- **Cell body (soma):** contains the nucleus and the life-support organelles.
- **Axon:** wire-like fiber that carries the outgoing signal from the soma toward the terminal buttons.
- **Myelin sheath:** fatty insulation around the axon that speeds neural impulses; produced by Schwann cells.
- **Nodes of Ranvier:** gaps in the myelin where the action potential “jumps,” speeding transmission on myelinated neurons.
- **Terminal buttons (axon terminals):** branched axon ends that store neurotransmitters in vesicles, ready for release.
- **Synapse (synaptic cleft):** the tiny gap between one neuron’s terminal button and the next neuron’s dendrite — the site of chemical communication between neurons.

Neural transmission is electrochemical: *electrical* within the neuron (dendrites, soma, axon, terminals) and *chemical* between neurons (neurotransmitters cross the synapse; electricity does not jump the gap).

Misconception: the myelin sheath does not decide whether a neuron fires — it only speeds conduction. Deterioration of myelin (as in multiple sclerosis) slows or blocks transmission.

Exam tip: know the direction of the signal. “Receives” points to dendrites; “sends” points to the axon and terminal buttons; the synapse is where neurotransmitters — never electricity — travel.

Unit 2 — Cognition

Memory: The Information-Processing Model

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working-memory

Memory: the persistence of learning over time through three processes — **encoding** (getting information in), **storage** (retaining it), and **retrieval** (getting it out).

Information-processing model: memory works like a computer — encode, store, retrieve. **Automatic processing** encodes incidental information unconsciously (space, time, frequency); **effortful processing** requires attention and conscious work.

Memory types:

- **Explicit (declarative) memory:** conscious memories of facts and experiences, easily described — **episodic memory** (personal experiences and events) and **semantic memory** (facts and general knowledge).
- **Implicit (nondeclarative) memory:** retention without conscious recollection — **procedural memory** for skills and operations (riding a bike).

- **Prospective memory:** remembering to perform planned actions at future moments (taking medication at 8 p.m.).

Neural basis: long-term potentiation (LTP) — the persistent strengthening of synapses based on recent activity patterns; receiving neurons become more efficient (more receptors). LTP is the neural basis of learning and memory.

Multi-store (Atkinson-Shiffrin) model: sensory memory, then short-term memory, then long-term memory.

- **Sensory memory:** immediate, brief recording of sensory information — **iconic memory** (visual, a few tenths of a second) and **echoic memory** (auditory, 3-4 seconds).
- **Short-term memory (STM):** briefly holds about 7 items before storage or loss (20-30 seconds without rehearsal).
- **Long-term memory (LTM):** relatively permanent, limitless storage.

Working memory model (Baddeley): an updated view of STM as ACTIVE processing of incoming and retrieved information, with a limited capacity for simultaneous processing:

- **Central executive:** directs attention and processing.
- **Phonological loop:** verbal/auditory information.
- **Visuospatial sketchpad:** visual/spatial information.
- **Episodic buffer:** integrates the streams (taught).

Levels of processing: retention depends on processing DEPTH — **structural** (physical features, shallowest), **phonemic** (sound), **semantic** (meaning — deepest, best recall).

Misconception: short-term memory and working memory are not synonyms on the exam — STM stores briefly; working memory actively manipulates information with named components.

Exam tip: map the scenario to the model: “recalling a phone number just long enough to dial” = STM; “mentally rotating furniture in a room” = visuospatial sketchpad; “knowing Paris is the capital of France” = explicit/semantic.